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CSA15 Cloud Computing and Big Data Analytics

CO1 AT2 - Digital Concept Map and Service Model Matrix

Individual Student Submission Template

Student and Faculty Details

Student Name	PAWAN KALYAN R	Register Number	192424116
Class / Section		Slot	C
Course Code	CSA1512	Course Title	Cloud Computing and Big Data Analytics
Capstone Title	Weather temperature forecasting using stacked RNNs		
Submission Date			
Faculty Name	Dr RAJESH BABU CHANDRAN	Employee ID	
Department			

Assessment Brief

Course Outcome	CO1 - Explain the evolution, architecture, characteristics, service models, and emerging paradigms of cloud computing.
Assessment Type	Individual concept mapping, service-model responsibility analysis, and capstone-cloud interpretation.
Assessment Focus	Each map must show key nodes, link labels, relationship meaning, and a logical hierarchy. The response must connect cloud fundamentals to the student's own perception of the capstone title.
Submission Mode	Editable DOCX + exported diagram evidence + GitHub repository link submitted in Google Classroom.
Permitted Tools	draw.io / diagrams.net, Lucidchart, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool.
Individual Rule	Every student must submit their own concept map, matrix, capstone-cloud mapping, and reflection. The capstone title may be shared by a team, but the assessment answer must not be common.
Total Marks	100 marks

Code of Conduct and Student Assurance

Academic Integrity Statement

This assessment must be completed individually. Students may discuss the capstone domain with their team, but the concept map, service model matrix, explanations, GitHub evidence, and reflection submitted here must be individually prepared.

Student Assurance	I confirm that this submission represents my own understanding of CO1 concepts and my individual interpretation of the selected capstone title.
Tool Use	I have used digital tools only for diagram preparation, organization, and presentation. I have not copied another student's diagram, matrix, or explanation.
Evidence	I have uploaded the required files to my GitHub repository and submitted the repository link / evidence link through Google Classroom.
Student Signature	
Date	

Procedure for Preparing the Digital Concept Map

Use the following procedure to prepare the diagram and upload evidence. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.

1. Open <https://app.diagrams.net/> or the desktop version of draw.io / diagrams.net.
2. Choose Device, Google Drive, or OneDrive as the save location according to availability.
3. Create a blank diagram and name the file as CSA15_CO1_AT2_RegisterNumber_Concept_Map.drawio.
4. Place the capstone title at the center or as a clearly visible anchor node.
5. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
6. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
7. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for capstone-specific elements.
8. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
9. Paste the exported diagram image or screenshot in this DOCX under Task 1.
10. Upload the DOCX, exported PNG/PDF, and .drawio file to GitHub, then submit the GitHub link in Google Classroom.

GitHub Submission Procedure

Step	Required Action	Expected Evidence
1	Create or use your individual GitHub account.	GitHub profile link.
2	Create a repository named CSA15-CO-Assessments-RegisterNumber or use the repository instructed by faculty.	Repository URL.
3	Create a folder named CO1_AT2 inside the repository.	Folder visible in repository.
4	Upload this completed DOCX, exported diagram PNG/PDF, and editable .drawio source file.	At least three evidence files uploaded.
5	Add a short README.md explaining the capstone title and files submitted.	README visible in the AT2 folder.
6	Submit the GitHub folder link in Google Classroom.	GCR submission with repository/folder link.

Assessment Questions and Student Response Sections

Important

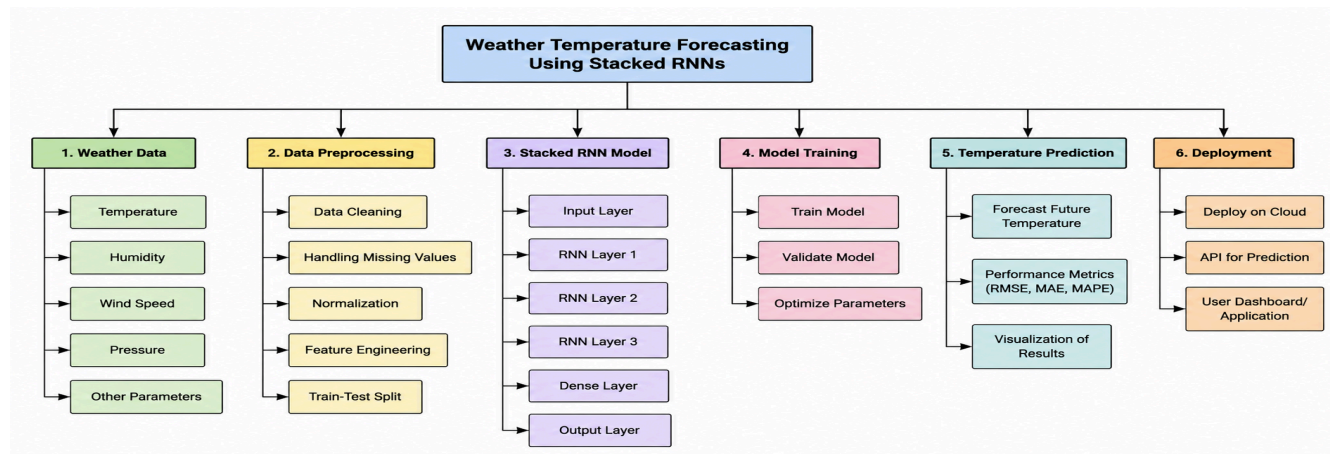
The capstone project may be developed by a team, but every response below must be written from the individual student's own perspective. Do not submit a common team answer.

Task 1 - Digital Cloud Concept Map

Create one integrated digital concept map for CO1. The map must show key nodes, link labels, relationship meaning, and a logical hierarchy. It must also connect the concepts to your selected capstone title from your individual perspective.

Mandatory Concept Group	Minimum Concepts / Nodes to Include
Cloud Fundamentals	Definition, characteristics, benefits, limitations, on-demand access, measured service.
Cloud Evolution and Drivers	Hardware evolution, internet software evolution, distributed systems, cost economics, mobility, remote access, big data growth.
Architecture and Deployment	Front-end, back-end, network layer, data center, management software, public/private/hybrid/community cloud.
Service Models and Responsibility	IaaS, PaaS, SaaS, XaaS, provider responsibility, user responsibility, shared responsibility.
Virtualization and Web/API Layer	Virtual machine, hypervisor, abstraction, resource pooling, multi-tenancy, web services, SOAP, REST, API request/response.
Capstone Connection	Your capstone users, app/module, cloud backend, database/storage, dashboard/analytics, scalability and elasticity idea.

Paste exported concept map image / screenshot here. Resize neatly within this white space.



Brief explanation of your concept map: explain the most important nodes, link labels, hierarchy, and capstone connection.

Root Node: Weather Temperature Forecasting Using Stacked RNNs.

- **Main Nodes:** Weather Data, Data Preprocessing, Stacked RNN Model, Model Training, Temperature Prediction, and Deployment.
- **Link Labels:** The arrows show the flow of data from collection → preprocessing → model building → training → prediction → deployment.
- **Hierarchy:** Top level is the project title, followed by six major modules, and then their related sub-processes.
- **Capstone Connection:** The concept map represents the complete machine learning pipeline for predicting future temperatures using Stacked RNNs and deploying the model for real-world use.

Task 2 - Service Model Matrix

Complete the matrices by making technical decisions for your capstone project. Select only the rows that are relevant to your capstone design. If a row is not used, write a short technical reason instead of leaving it blank.

Your answer should show how responsibility shifts between cloud provider and student development team, and how the chosen services support the capstone workflow, API design, storage, security, scalability, and evidence submission.

Task 2A - Service Model Responsibility Matrix

Service Model	Capstone Role / Module	Provider Responsibility	Student Responsibility	Evidence to Submit
IaaS	Virtual machine for model deployment	Provides virtual servers, storage, and networking	Configure VM, install Python, TensorFlow, deploy application	VM screenshot, deployment configuration
PaaS	Model training and API hosting	Manages runtime, web server, scaling, and updates	Develop RNN model, REST API, and deploy application	API URL, deployment screenshot
SaaS	Data visualization and dashboard	Provides dashboard platform and reporting tools	Upload prediction results and create visualizations	Dashboard screenshots
XaaS / Specialized Service	Weather data API	Supplies real-time and historical weather data	Fetch data through API and preprocess for training	API response screenshot, sample dataset

Task 2B - Capstone Cloud Service Decision Matrix

Cloud Layer / Function	Selected Tool or Service	Data / Resource Handled	API or Integration Path	Security / Scaling Note
Client access and UI	Streamlit	User input and prediction display	HTTP Request → Streamlit App	HTTPS enabled
API / backend logic	Flask REST API	Temperature prediction requests	REST API	Stateless API supports scaling
Authentication and identity	Firebase Authentication	User login details	Firebase SDK	Secure authentication and access control
Database / structured records	MySQL	Weather data and prediction records	Flask → MySQL Connector	User authentication and backups
File / object storage	Google Cloud Storage	Trained model (.h5), datasets	Cloud Storage API	Secure cloud storage and scalability
Monitoring / logs / usage metrics	Google Cloud Logging	API logs and application logs	Cloud Logging API	Continuous monitoring
Analytics / reporting output	Power BI / Tableau	Prediction reports and charts	Database → BI Tool	Read-only dashboards

Technical justification: Explain why the selected cloud service pattern fits your capstone project's users, data, API flow, scalability, and security needs.

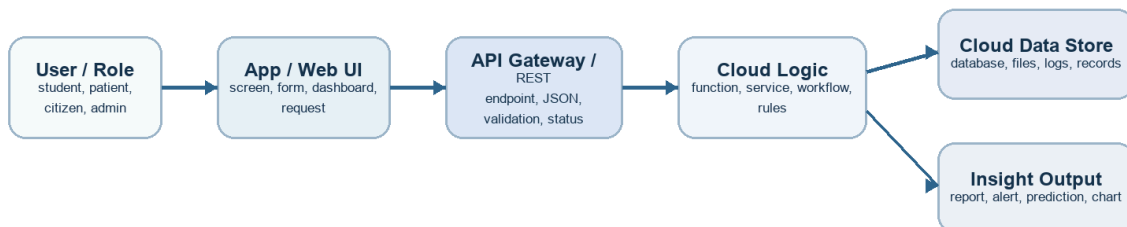
The PaaS-based architecture is the best choice for this capstone because it simplifies deployment and automatically manages the application environment. Flask APIs connect the Streamlit interface with the Stacked RNN model, while MySQL stores weather and prediction data. Google Cloud Storage securely stores datasets and trained models. Firebase Authentication protects user access, and Cloud Logging monitors application performance. This architecture provides high scalability, secure API communication, reliable storage, and easy deployment, making it suitable for accurate weather temperature forecasting using Stacked RNNs.

Task 3 - Capstone Cloud Flow Mapping

Study the sample architecture-flow template below. Redraw it in draw.io / diagrams.net using your own capstone architecture. Replace every generic label with your actual user role, screen, API endpoint, backend logic, cloud storage, analytics output, security control, and scalability or elasticity decision.

Sample Capstone Cloud Flow Template

Replicate this structure using your own capstone screens, APIs, cloud services, data stores, and outputs.



Cross-cutting cloud decisions to label in your own diagram

Authentication / IAM

Who is allowed to access each API and data item?

Scalability

Which component must handle growing users or records?

Elasticity

Which component can expand or shrink based on demand?

Monitoring

What logs, usage metrics, or errors will be observed?

Security

Where are validation, privacy, backup, and access control applied?

Student instruction: redraw this flow in draw.io/diagrams.net and replace every label with your own capstone architecture.

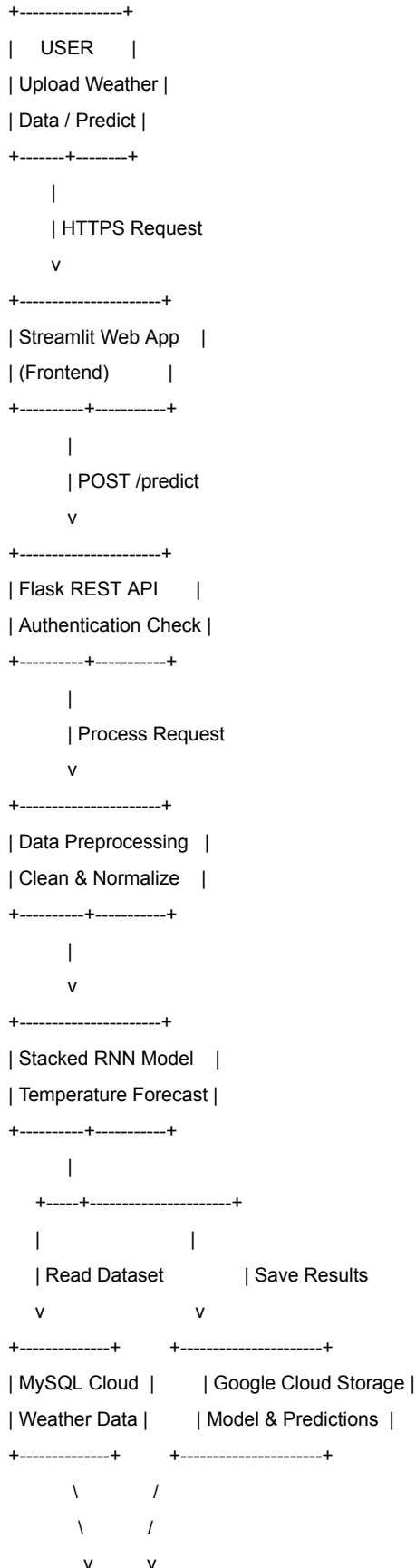
Student Drawing Requirement

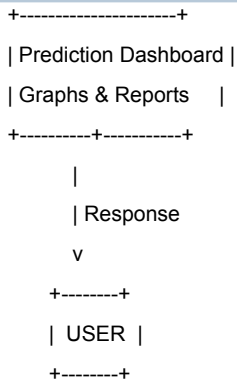
Do not copy the sample as it is. Use it as a layout model, then create your own capstone-specific flow diagram with meaningful labels, arrows, data movement, API names, and cloud service choices.

Flow Component	Student Entry
User role and action	User uploads weather data and requests temperature prediction
Frontend / app screen	Streamlit Web Application
API endpoint and method	POST /predict (Flask REST API)
Backend processing rule	Preprocess data → Load Stacked RNN model → Generate temperature prediction
Cloud database / storage operation	Read weather dataset from MySQL → Load trained model from Google Cloud Storage → Store prediction results
Response, alert, report, or dashboard output	Predicted temperature, performance graph, dashboard report
Security, scalability, and elasticity labels added in diagram	HTTPS, Firebase Authentication, Auto Scaling, Cloud Storage, Load Balancer

Paste your redrawn capstone cloud-flow diagram here.

WEATHER TEMPERATURE FORECASTING USING STACKED RNNs (CLOUD)





Security: HTTPS • Firebase Authentication

Scalability: Auto Scaling • Load Balancer

Storage: MySQL • Google Cloud Storage

Brief explanation: explain one complete request-response path from user action to cloud response.

When the user uploads weather data through the Streamlit web application, the request is sent to the Flask REST API using the POST /predict endpoint. The backend authenticates the user, preprocesses the weather data, and sends it to the Stacked RNN model to forecast future temperature. The model reads historical data from MySQL, loads the trained model from Google Cloud Storage, and stores the prediction results. Finally, the predicted temperature and performance graphs are displayed on the dashboard and returned securely to the user through HTTPS.

Evaluation Rubric - 100 Marks

Criteria	Marks	Expected Standard
Concept identification	20	Major CO1 concepts are accurately identified: cloud fundamentals, characteristics, evolution, service models, deployment models, virtualization, web services, SOAP/REST, APIs, elasticity, and scalability.
Relationship mapping	20	Links between concepts are accurate, meaningful, and labeled. The map should explain how concepts depend on, enable, consume, provide, scale, or control one another.
Hierarchy and organization	15	Concepts are arranged in a logical hierarchy with clear grouping, readable flow, and no disconnected keyword clusters.
Technical relevance	15	The map strongly reflects cloud course content and avoids decorative or generic non-technical content.
Service model matrix	15	IaaS, PaaS, SaaS, and XaaS are compared with provider responsibility, user responsibility, examples, capstone usage, and risks or limitations.
Capstone alignment and individual reflection	10	The student's own capstone interpretation is visible in the concept map, flow mapping, justification, and reflection. Team-common answers are avoided.
Visual clarity and digital evidence	5	Diagram is neat, readable, exported properly, and supported with GitHub evidence including source and exported files.

Faculty Evaluation Record

Evaluator Name		Marks Awarded	
Observation		Signature	